MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG4DSCBOT200** Course Title: **ARCHEGONIATES**

Time: 1.5 Hrs **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part **A**: **Short answers**
Answer 10 questions
Each question carries 1 mark

	Questions	Taxonomy	CO
1	Name the first cell of sporophytic generation.	Remember	1
2	How do the sex organs in archegoniates differ from that of algae?	Understand	1
3	Name the two types of rhizoids in *Riccia.*	Remember	3
4	What are pseudoelators?	Understand	3
5	Name the conducting tissues found in the central axis of *Pogonatum.*	Remember	3
6	Is *Psilotum* homosporous or heterosporous?	Remember	3
7	What is ligule and where is it located?	Understand	3
8	Which pteridophyte is widely used as a biofertilizer?	Remember	4
9	Why do gymnosperms fail to produce fruits?	Understand	1
10	Name the three classes of gymnosperms proposed by Sporne.	Remember	2
11	What is the symbiotic relationship found in *Pinus* roots?	Remember	3
12	List out two common ornamental gymnosperms.	Remember	4

Part **B**:
Short Essay Type Questions
Answer 6 Questions
Each question carries 5 marks

13	Discuss the challenges faced by early land plants for the transition from water to land and the adaptations they evolved.	Understand	1
14	Explain the unique features of sporophyte of *Anthoceros*	Understand	3
15	Write a short note on the synangium of *Psilotum*.	Understand	3
16	Analyze how heterospory in *Selaginella* provides an evolutionary advantage in terrestrial environments.	Analyze	3
17	Discuss different stelar types in pteridophytes.	Understand	3
18	Describe the xerophytic features of *Cycas* leaflet.	Understand	3
19	Comment on structure of *Pinus* ovule.	Understand	3
20	Summarize the economic importance of gymnosperms.	Understand	4

Part **C**:

Essay Type Questions

Answer any one question

Question carries 10 marks

21	Assess the ecological and economic importance of pteridophytes.	Evaluate	4
22	Write an essay on general characteristics of gymnosperms.	Understand	1